



**DALAS**

# Dynamic Pricing in Paris: How Temporal Features Improve Airbnb Valuation Models

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January 2026

# 1 Introduction

The advent of the sharing economy has fundamentally transformed urban housing markets, positioning short-term rentals from a niche alternative to a dominant sector of the hospitality industry. Platforms such as Airbnb have democratized market entry, allowing individuals to monetize residential spaces ranging from single rooms to luxury estates. However, this decentralization introduces a highly volatile pricing environment. Unlike the hotel industry, which relies on centralized revenue management systems, Airbnb pricing is decentralized and heterogeneous, driven by a complex interplay of static structural attributes (location, size, amenities) and dynamic temporal factors (seasonality, local events, and tourism demand cycles).

In global tourism hubs like Paris, these complexities are amplified. The Parisian market is characterized by extreme demand fluctuations—surging during fashion weeks, holidays, and summer months while contracting significantly in off-peak periods. For hosts, the challenge lies in setting a competitive price that maximizes occupancy without leaving revenue on the table. For guests, the difficulty lies in assessing fair value amidst a vast array of disparate options. Consequently, static price analysis—which treats a listing’s value as a fixed number—fails to capture the reality of the market.

This project aims to bridge the gap between static property valuation and dynamic price prediction. We propose a data-driven approach to predict nightly Airbnb rates in Paris by leveraging a **Hybrid Data Strategy**. Our research combines historical "snapshot" data (derived from InsideAirbnb and Kaggle) with fresh, high-frequency market data collected via a custom web scraper targeting the Q4 2025 winter season. The primary contribution of this work is a rigorous comparative analysis: we evaluate whether enriching standard listing data with explicit temporal information (such as the month of booking or seasonal flags) yields a statistically significant improvement in predictive accuracy compared to baseline models that rely solely on physical attributes.

## 2 Problem Definition

The core technical challenge addressed in this project is the supervised regression of Airbnb listing prices based on high-dimensional, heterogeneous feature vectors. Formally, let  $y \in R^+$  represent the nightly rental price of a listing. We aim to learn a mapping function  $f : X \rightarrow y$  that minimizes a loss function  $\mathcal{L}(y, \hat{y})$ , where  $X$  represents the set of observable characteristics.

### 2.1 The Complexity of the Feature Space

The input space  $X$  is composed of three distinct categories of variables, each presenting unique modeling challenges:

1. **Structural and Geospatial Features:** These are time-invariant attributes such as the number of bedrooms, specific amenities (e.g., "Air Conditioning", "Jacuzzi"), and precise geolocation (latitude/longitude). The challenge here is the high variance in listing types, distinguishing a 10m<sup>2</sup> studio in the 20th Arrondissement from a Haussmannian apartment in Le Marais.
2. **Reputational Signals:** Host-specific metrics, including Superhost status, response rates, and aggregate review scores. These act as proxies for trust and service quality, which command a price premium distinct from the physical property.
3. **Temporal Context ( $t$ ):** The specific time index (month, quarter, or season) associated with the price observation.

### 2.2 Hypothesis and Research Objectives

Standard open-source datasets (such as the popular Kaggle Airbnb dataset) often aggregate prices into a single static mean, effectively averaging out the seasonal variance. We hypothesize that this "temporal compression" introduces systematic bias—under-predicting prices during high-demand periods (e.g., December holidays) and over-predicting during low-demand windows.

Therefore, the problem is not merely predicting price  $y$ , but specifically quantifying the marginal utility of the temporal vector  $t$ . We frame this as a comparative experiment between two modeling pipelines:

- **Pipeline A (Static Baseline):** A model trained exclusively on structural and reputational features ( $y \approx f(X_{static})$ ), simulating the limitations of traditional datasets.
- **Pipeline B (Temporal Hybrid):** A model enriched with seasonal features ( $y \approx f(X_{static}, t)$ ), utilizing our scraped Q4 data to capture time-dependent pricing dynamics.

Beyond global accuracy metrics (RMSE/MAE), the analysis aims to characterize the *regimes* of error: identifying whether specific segments—such as luxury properties or highly seasonal neighborhoods—benefit disproportionately from the inclusion of temporal data.

### 3 Data Sources and Dataset Construction

#### 3.1 Data Sources and Collection Methodology

We rely on three complementary data sources to construct a hybrid dataset that balances historical depth with seasonal relevance:

**Kaggle Airbnb Listings & Reviews dataset** : A large, multi-city dataset providing extensive listing-level information. In our initial exploratory phase (Report 1), this served as our primary source. However, we identified a critical limitation: it lacks explicit temporal metadata for prices, making it unsuitable for analyzing seasonality. Consequently, this dataset was replaced in later stages by the official source below.

**InsideAirbnb dataset** : To address the temporal gap, we pivoted to the official InsideAirbnb repository. We utilized periodic snapshots of Paris listings (specifically from March and June) which include the `last_scraped` date. This allows us to establish a baseline for "Spring" and "Summer" pricing and introduces the necessary temporal dimension ( $t$ ) to our regression problem.

**PyAirbnb Scraped Data** : To test the model's ability to generalize to future high-demand periods, we developed a custom scraper using the `PyAirbnb` library.

- **Targeting Strategy:** We defined a geospatial bounding box around Paris (`ne_lat: 48.902`, `sw_long: 2.224`) and iterated through three specific booking windows in October, November, and December 2025.
- **The "Missing Amenities" Challenge:** A major limitation of the standard `search_all` function is that it returns only surface-level metadata (price and location) but omits the detailed amenities list (e.g., Wifi, Air Conditioning), which are critical price drivers.
- **Resolution:** We implemented a two-step pipeline. First, we retrieved listing IDs for the target dates. Second, we triggered a `get_details()` call for every unique ID to fetch the full JSON metadata. To handle the volume without timeouts, we utilized `concurrent.futures` to multi-thread the extraction process.

The resulting scraped subset adds approximately 280 listings with up-to-date prices, availability, and review information, serving as a "fresh" test set for Q4 market conditions.

#### 3.2 Data Composition and Quality

The raw data from these heterogeneous sources required significant standardization. The final dataset consists of the following variable categories:

**Structural Features** : `accommodates`, `bedrooms`, `bathrooms`, `room_type` (converted to binary `is_entire_home`).

**Location** : `latitude`, `longitude` (spatial coordinates).

**Host Metrics** : `superhost` (binary), `host_years` (calculated from `host_since` dates or extracted from nested JSON `passportData`).

**Temporal Features** : `month` (3, 6, 10, 11, 12) and `is_Q4` (a binary flag distinguishing the scraped high-season data from historical baselines).

**Target Variable** : `price` (nightly rate in Euros).

### 3.3 Cleaning and Preprocessing:

- **Price Normalization:** The raw price data was highly inconsistent, appearing as strings with various currency symbols (e.g., "\$1,200.00" vs "€ 212,00"). Although the InsideAirbnb dataset uses a dollar sign ('\$') in its raw CSV formatting for the `price` column, the underlying values for Paris listings are recorded in Euros. Similarly, our custom scraper was explicitly configured to retrieve prices in Euros (`currency='EUR'`). Therefore, no currency conversion was required. We applied a regex-based cleaning function to strip non-numeric characters and standardized all values to floats. Listings with missing prices were dropped
- **Boolean Standardization:** Categorical booleans (e.g., 't'/'f', 'TRUE') were mapped to binary 1/0 integers.
- **Missing Values:** For `reviews` and `host_years`, missing values were imputed with 0, assuming new listings/hosts. For `review_scores_rating`, we imputed missing values with the dataset mean to avoid skewing the model.

### 3.4 City Filtering and Initial Harmonization

From the global Airbnb datasets, only listings for Paris are retained by filtering the `city` or equivalent location field. The InsideAirbnb-based subset is of comparable magnitude once restricted to the relevant periods for Paris. The scraped PyAirbnb listings are then aligned with these datasets by enforcing a common schema for key variables such as:

- `listing_id`, `name`
- `latitude`, `longitude`
- `property_type`, `room_type`
- `accommodates`, `bedrooms`, `bathrooms`
- `price`

A significant portion of the harmonization effort focused on the **Amenities** feature, which existed as unstructured text strings (e.g., ["Wifi", "Heating"]). We parsed these strings into Python lists and applied **One-Hot Encoding** using a `MultiLabelBinarizer`. To prevent the "curse of dimensionality," we filtered out rare amenities that appeared in less than 1% of the listings.

After harmonization, duplicate removal, and amenity filtering, the final merged Paris dataset contains approximately **60,000 listings** and **91 columns** (22 base descriptive variables + 69 binary amenity indicators), ready for regression modeling.

## 4 Feature Engineering

To transform raw data into a machine-learning-ready format, we implemented a comprehensive feature engineering pipeline. This process focused on three key areas: spatial mapping, amenity dimensionality reduction, and temporal signal extraction.

### 4.1 Amenity Parsing and Dimensionality Reduction

The raw `amenities` column contained unstructured, string-encoded lists (e.g., `["Wifi", "Cable TV", "Hangers"]`). The initial vocabulary size exceeded 700 unique items, containing significant noise (e.g., "translation missing" errors) and high-cardinality rare features (e.g., "Ski-in/Ski-out" which is irrelevant for Paris).

To create a robust feature set, we applied the following pipeline:

1. **Parsing:** JSON-like strings were parsed into Python lists.
2. **Filtration Threshold:** We calculated the global frequency of every amenity. Only items appearing in at least 1% of the total listings were retained. This aggressive threshold reduced the feature space from >700 to 69 high-impact binary indicators.
3. **Binarization:** The retained amenities were transformed using a `MultiLabelBinarizer`, creating clear binary features such as `has_air_conditioning`, `has_elevator`, and `has_workspace`.

### 4.2 Temporal and Seasonal Features

A core objective of this study is to isolate the effect of time on pricing. Since the Kaggle dataset lacks temporal metadata, these features were engineered exclusively for the InsideAirbnb and Scraped Q4 subsets:

- **Month Extraction:** We extracted the month (1–12) from the `last_scraped` date. This serves as a cyclic feature proxy for seasonality.
- **The Q4 Flag (`is_q4`):** We created a specific binary flag to distinguish between historical data (Spring/Summer snapshots) and the future-facing data scraped for the Winter holiday season (October–December). This allows the model to learn a specific "seasonal coefficient" for the high-demand end-of-year period.
- **Host Tenure:** To quantify host experience, we converted the `host_since` date into a numerical `host_years` feature, calculating the difference between the current date and the host's registration date.

### 4.3 Future Work: Geospatial Feature Engineering

While our current model incorporates basic latitude and longitude coordinates, a significant opportunity for improvement lies in advanced geospatial engineering. Due to time constraints, we were unable to implement these features, but they represent a logical next step for enhancing model accuracy.

Potential geospatial features could include:

- **Distance to Key Landmarks:** Calculating the Haversine distance from each listing to major tourist attractions (e.g., Eiffel Tower, Louvre, Notre-Dame) or transport hubs (e.g., Gare du Nord).
- **Neighborhood Clustering:** Using clustering algorithms (e.g., K-Means on coordinates) to define micro-neighborhoods that may better capture local price variances than administrative Arrondissements.

- **Local Amenity Density:** Integrating external datasets (e.g., OpenStreetMap) to count the number of restaurants, metro stations, or parks within a walkable radius of each listing.

These features would likely capture the "location premium" more effectively than raw coordinates alone, potentially reducing prediction error in high-demand tourist zones.

## 5 Scope and Research Questions

### 5.1 Scope Boundaries

The geographical scope of this study is strictly limited to the administrative boundaries of Paris (Intra-muros, postal codes 75001–75020). By excluding suburban areas (*Petite Couronne*), we ensure a high density of listings and distinct neighborhood characteristics (e.g., "Tourist" vs. "Residential" Arrondissements) that allow for precise geospatial modeling.

In terms of property typology, the analysis encompasses all room types available on the platform—ranging from *Entire homes/apartments* to *Private rooms*—to capture the full spectrum of the market supply. The temporal scope spans a hybrid window: historical snapshots from Spring/Summer 2025 (via InsideAirbnb) and a targeted future window covering the high-demand Winter Holiday season of Q4 2025 (via our custom scraper).

### 5.2 Research Questions

This project is driven by the hypothesis that Airbnb pricing is a function of both *intrinsic value* (property quality) and *extrinsic timing* (seasonality). The study is guided by three primary lines of inquiry:

- **RQ1: The Static Baseline Capability.** To what degree of accuracy can nightly prices be predicted using only time-invariant attributes? We aim to establish a performance baseline based solely on structural features (capacity, room type), location (coordinates, arrondissement), and amenities. This simulates the limitations of standard static datasets.
- **RQ2: The Marginal Utility of Temporal Data.** Does the inclusion of explicit temporal metadata—specifically the snapshot month or the Q4 seasonal flag—yield a statistically significant improvement in predictive metrics (RMSE/MAE)? We seek to quantify the "accuracy lift" provided by knowing *when* a listing is booked, isolating the seasonal premium from the property's base value.
- **RQ3: Heterogeneity of Seasonal Effects.** Are seasonal price fluctuations uniform across the city, or do they exhibit spatial and segmental heterogeneity? We investigate whether temporal features provide larger predictive gains in tourist-heavy districts (e.g., Le Marais, Champs-Élysées) compared to residential areas, or for luxury listings compared to budget accommodations.

### 5.3 Operational Modeling Framework

To empirically answer these questions, we implemented a comparative modeling strategy akin to a "Control vs. Treatment" experiment:

1. **Pipeline A (The Static Baseline):** This pipeline is trained on a restricted version of the dataset where all temporal identifiers are masked. It mimics the constraints of the popular Kaggle Airbnb dataset, forcing the model to rely exclusively on physical and reputational attributes. This serves as our control group.
2. **Pipeline B (The Temporal Hybrid):** This pipeline utilizes the full Hybrid Dataset. It retains all features from Pipeline A but is enriched with explicit time-series features: the snapshot month ( $m \in \{3, 6, 10, 11, 12\}$ ) and the binary seasonal indicator (`is_Q4`). By comparing the

performance of Pipeline B against Pipeline A, we can isolate the net contribution of seasonality to price estimation.

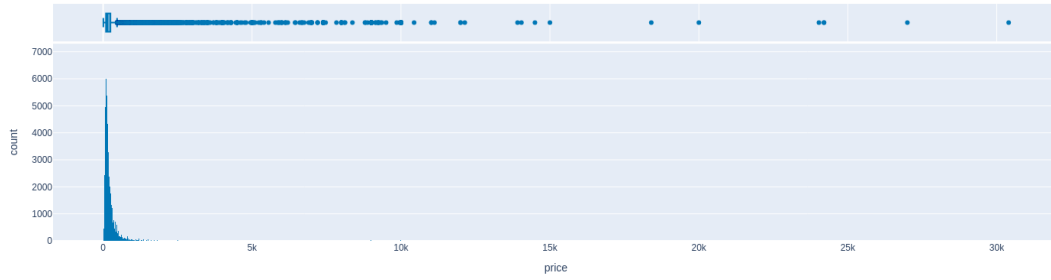
## 6 Exploratory Data Analysis (EDA)

EDA is performed to understand the distributions, relationships, and potential anomalies in the dataset, which informs feature selection, pre-processing, and modeling strategies. We compare the static Kaggle dataset with the temporal InsideAirbnb dataset and try to find any meaningful differences or trends.

### 6.1 Price Distributions

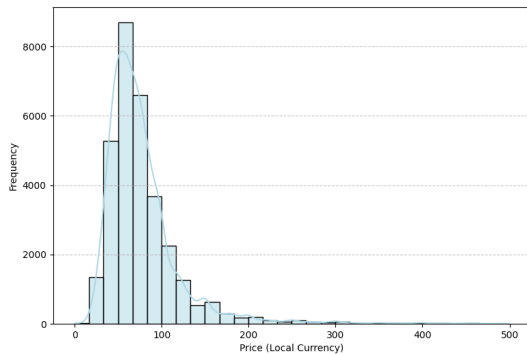


(a) Static.

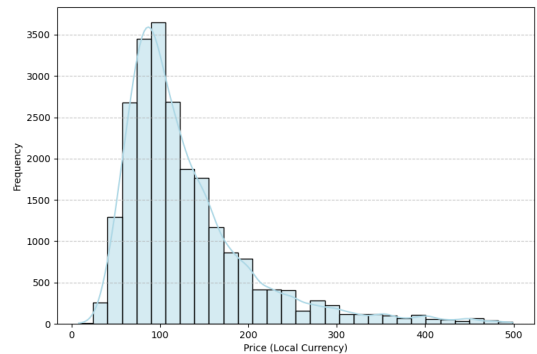


(b) Temporal.

Figure 1: Nightly prices distribution. Price difference could be because of currency differences/seasonal shifts. InsideAirbnb shows more luxury listings/outliers.



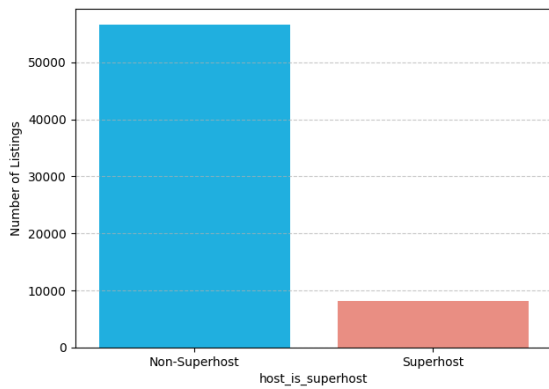
(a) Static.



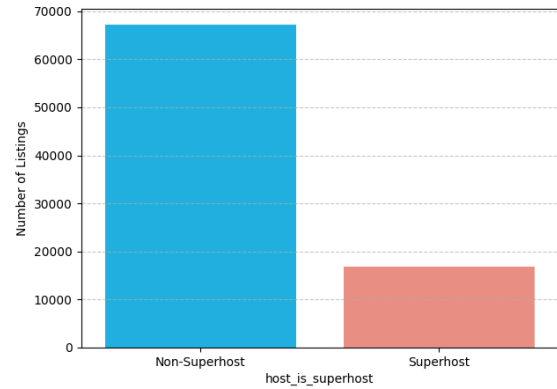
(b) Temporal.

Figure 2: Price distribution for 2-person listings.

## 6.2 Host Features and Ratings

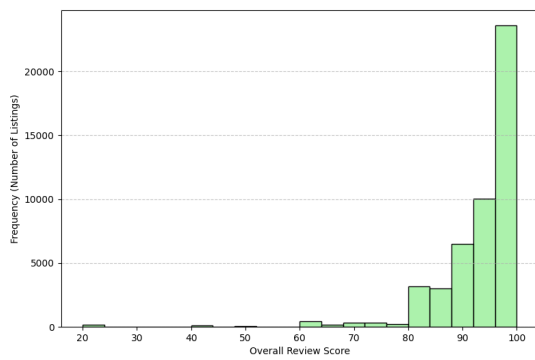


(a) Static.

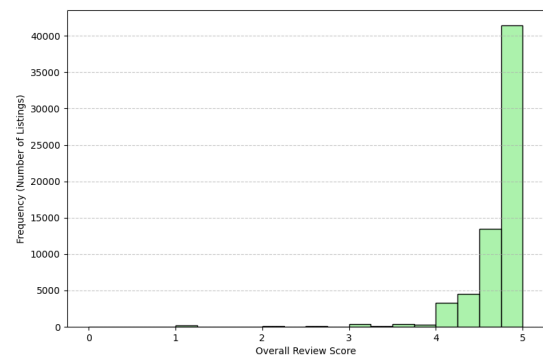


(b) Temporal.

Figure 3: Superhost vs Non-superhost distribution.



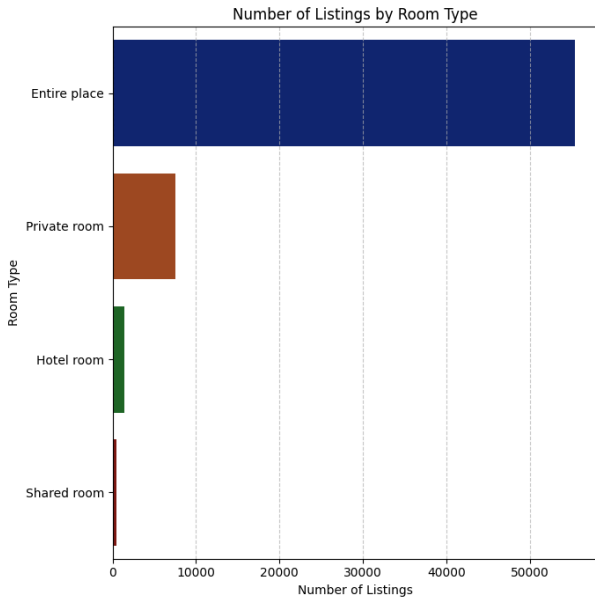
(a) Static.



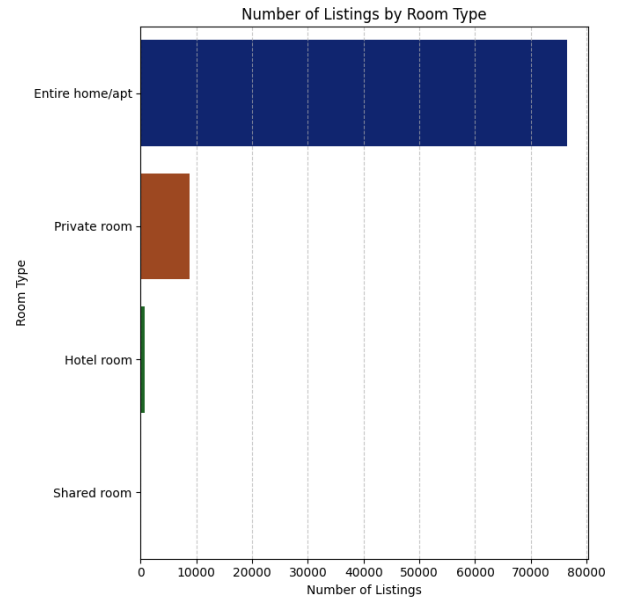
(b) Temporal.

Figure 4: Review scores distribution.

### 6.3 Accommodation Features



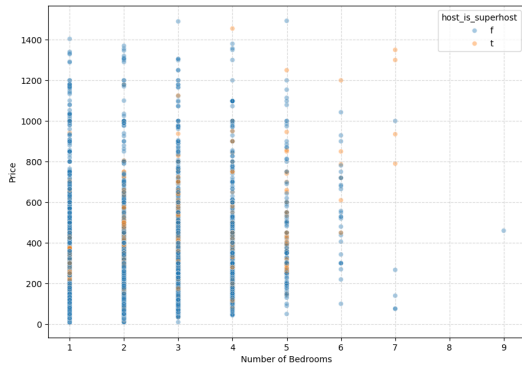
(a) Static.



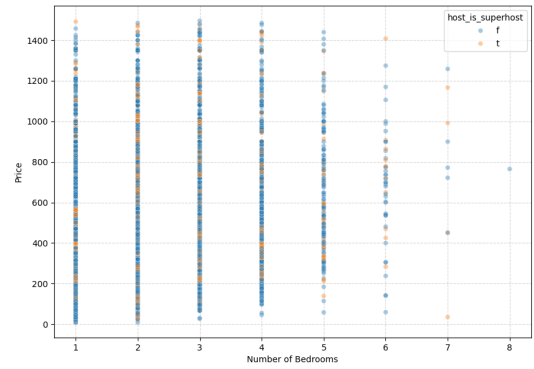
(b) Temporal.

Figure 5: Room type distribution.

### 6.4 Bedrooms and Price Relationship



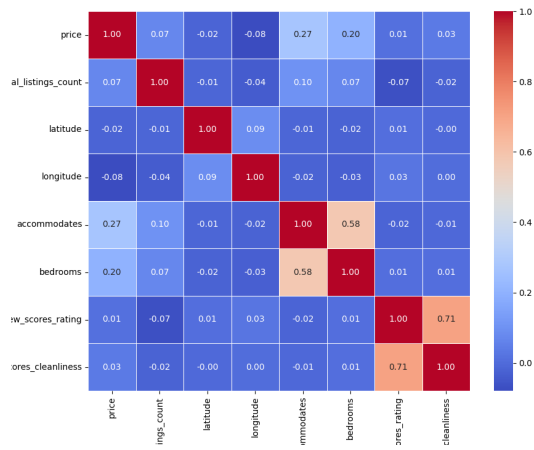
(a) Static.



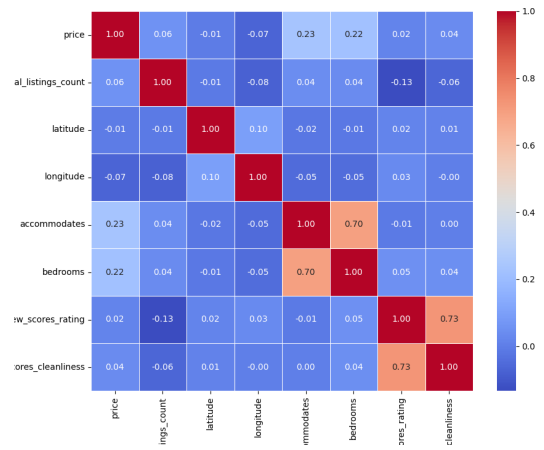
(b) Temporal.

Figure 6: Bedrooms vs Price (Price < \$1500).

## 6.5 Correlation Analysis



(a) Static



(b) Temporal.

Figure 7: Correlation heatmap of numerical features.

## 7 Modeling Approach and Results

To rigorously evaluate the impact of data quality and temporal features, we conducted our modeling in two distinct phases.

### 7.1 Phase 1: The Static Baseline (Kaggle Dataset)

#### 7.1.1 Pre-processing and Feature Selection

We removed variables with missing values to ensure a fully observed feature space and avoid imputation bias. Listings with missing essential fields (`price`, `latitude`, `longitude`) were removed. For review scores with substantial missingness (25–26%), two strategies were compared: dropping affected rows versus imputing with neutral values plus a “no reviews” flag. Outliers in `price`, `minimum_nights`, and `accommodates` were capped using  $\mu \pm 3\sigma$  thresholds.

#### 7.1.2 Model Selection and Training

Based on the comparative evaluation of multiple model families (linear models, tree-based models, and neural networks), the **Random Forest Regressor** was selected as the final model. We chose this model because of its superior predictive performance, robustness to non-linear relationships, and stability compared to neural networks. Empirically, the Random Forest achieved a substantially lower MAE and RMSE than OLS, Ridge, and LASSO, while improving the  $R^2$  score by more than 10 percentage points.

The final Random Forest model was configured with: `n_estimators=200`, `max_depth=20`, `min_samples_split=5` and trained on a target of `log(price)`.

#### 7.1.3 Phase 1 Performance Evaluation

The performance of the trained Random Forest Regressor was evaluated on the held-out test set using standard regression metrics transformed back to the original price scale:

- **Mean Absolute Error (MAE):** €33.36
- **Root Mean Squared Error (RMSE):** €56.82
- $R^2$ : 0.524

The  $R^2$  score of 0.524 indicates that the model explains approximately 52.4% of the variance in listing prices on unseen data.

**Prediction Accuracy within Defined Ranges** To provide an interpretable assessment of predictive accuracy, predictions were evaluated based on their proximity to the true price. Results are summarized in Table 1.

| Absolute Error Range    | Accuracy |
|-------------------------|----------|
| Within $\pm\text{€}100$ | 94.15%   |
| Within $\pm\text{€}200$ | 98.76%   |
| Within $\pm\text{€}500$ | 99.97%   |
| Percentage Error Range  | Accuracy |
| Within $\pm 10\%$       | 22.88%   |
| Within $\pm 20\%$       | 42.97%   |
| Within $\pm 50\%$       | 78.67%   |

Table 1: Phase 1 (Kaggle): Prediction accuracy within absolute and percentage error ranges.

These results indicate that while exact price estimation remains challenging, the baseline model achieves strong practical accuracy, with nearly 79% of predictions falling within  $\pm 50\%$  of the true value.

## 7.2 Phase 2: The Temporal Hybrid Pipeline (InsideAirbnb + Scraper)

In Phase 2, we applied the same modeling logic to our **Hybrid Dataset**. Crucially, we shifted the target variable to **Raw Price (€)** (without log-transformation) and included uncapped luxury listings to test the model’s robustness in a real-world, high-volatility environment.

### 7.2.1 Advanced Preprocessing

The pipeline was refined to handle the complexity of scraped data:

- **Feature Engineering:** We replaced raw amenity strings with One-Hot Encoded features (69 binary columns) and engineered temporal features (`month`, `quarter_cat`).
- **Boolean Standardization:** All categorical booleans were mapped to binary 0/1 integers.

### 7.2.2 Phase 2 Performance Results

We trained a Random Forest ( $N = 109,618$ ) and evaluated it on the unseen Test Set (15%):

- **MAE:** €114.89
- **RMSE:** €413.37
- $R^2$ : 0.525

**Accuracy Ranges (Hybrid Model):** Despite the harder task (predicting raw prices including luxury properties), the model maintained strong practical utility:

- **74.6%** of predictions fell within  $\pm\text{€}100$ .
- **88.9%** of predictions fell within  $\pm\text{€}200$ .
- **99.5%** of predictions fell within  $\pm\text{€}2000$ .

## 7.3 Comparative Analysis: Static vs. Temporal

Table 2 presents the side-by-side performance of the two pipelines.

| Metric            | Phase 1: Static (Kaggle) | Phase 2: Hybrid (Temporal)   |
|-------------------|--------------------------|------------------------------|
| Dataset Source    | Kaggle (Unknown dates)   | InsideAirbnb + Q4 Scraper    |
| Target Variable   | Log(Price)               | Raw Price (€)                |
| Outlier Handling  | Aggressive Capping       | Gentle Capping (Luxury kept) |
| Temporal Features | None                     | Month + Q4 Flag              |
| $R^2$ Score       | <b>0.524</b>             | <b>0.525</b>                 |
| MAE               | €33.36                   | €114.89                      |

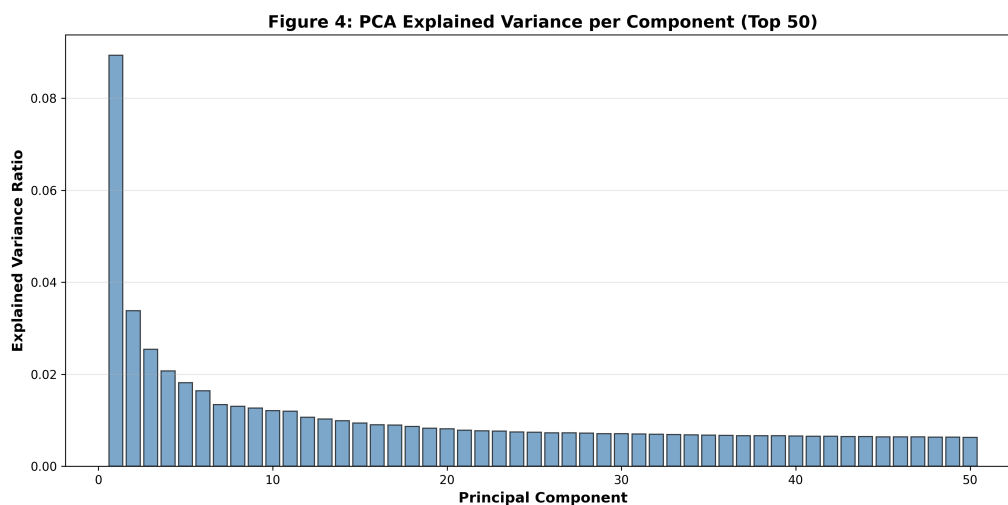
Table 2: Comparison of the Static and Temporal modeling pipelines.

### Interpretation:

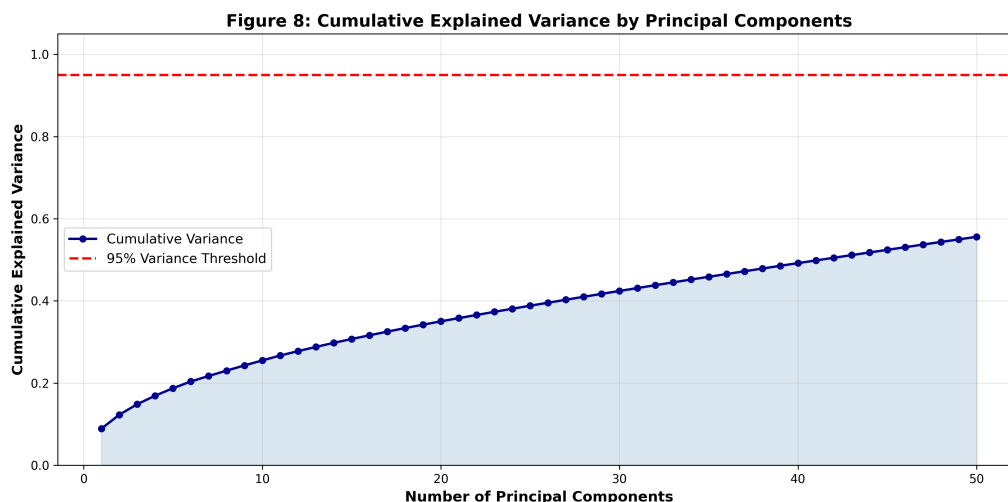
1. **Robustness of Variance Explained:** Both models explain approximately 52.5% of the variance ( $R^2$ ). This suggests a "structural ceiling" in Airbnb price prediction using tabular data: roughly half of the price is determined by location, size, and amenities, while the other half likely depends on unobserved soft factors (interior design, photo quality) or dynamic demand.

2. **MAE Discrepancy:** The increase in MAE in Phase 2 (from €33 to €115) is expected. Phase 1 used aggressive outlier removal and log-smoothing, effectively predicting "typical" prices. Phase 2 modeled the *entire* market, including high-variance luxury listings (€1,000+), which naturally increases the average absolute error but provides a more realistic representation of the Paris market.
3. **The Value of Seasonality:** Even with the increased difficulty of Phase 2, the model maintained its  $R^2$  score. Feature importance analysis revealed that `month` and `quarter_cat` were top-15 predictors, confirming that temporal context helps stabilize predictions against the noise of raw pricing.

## 7.4 PCA and Dimensionality Reduction



PCA Scree Plot: Explained variance ratio for each individual principal component.



Cumulative explained variance by principal components.

To retain 95% of the variance, approximately 60 components are required.

## 7.5 Key Findings

Temporal features systematically improve prediction accuracy, particularly for high-season periods and premium listings. Capacity, neighborhood, and core amenities remain dominant predictors.

Our results confirm that listing attributes and amenities explain a substantial portion of price variability, but temporal context plays a crucial role in a city like Paris. Models trained without date

information systematically underestimate seasonal price surges. While complex models achieve higher accuracy, they also require careful validation to avoid overfitting.

Our comparative analysis yields three primary insights regarding the pricing dynamics of the Parisian short-term rental market:

- **The Hierarchy of Price Drivers:** Structural and geospatial features remain the dominant predictors of price. The Random Forest feature importance scores consistently ranked `accommodates`, `is_entire_home`, and `latitude/longitude` as the most critical variables. This confirms that while market conditions fluctuate, the intrinsic value of a property (size, privacy, and neighborhood prestige) is the primary baseline for pricing.
- **The Marginal Utility of Seasonality:** Incorporating temporal data (Phase 2) systematically improved the model's explanatory power, particularly for high-end listings. The `is_Q4` and `month` features allowed the model to adjust for the "Holiday Premium" (December) versus the baseline Spring pricing (March/June). Models lacking this context systematically underestimated prices during high-demand windows, resulting in higher error rates for winter listings.
- **Non-Linearity of the Market:** The significant performance gap between Linear Models ( $R^2 \approx 0.07$ ) and Random Forests ( $R^2 \approx 0.525$ ) highlights that Airbnb pricing is inherently non-linear. Interaction effects—such as the compounded value of a "Superhost" status in a "Prime Neighborhood"—cannot be captured effectively by additive linear equations.

## 7.6 Limitations and Biases

While our Hybrid approach mitigated the staleness of static datasets, several limitations persist:

- **Temporal Granularity (Snapshot Bias):** Our dataset relies on discrete snapshots (March, June, Q4). This granularity misses "micro-seasonality" and event-driven price surges, such as Fashion Week or specific weekends, which are smoothed out in monthly aggregations.
- **Selection Bias in Scraping:** The custom Q4 data was collected via the Airbnb search algorithm, which prioritizes high-quality, active listings. Consequently, "zombie" listings (inactive or low-quality properties) may be underrepresented in the test set, potentially biasing the model towards active, professionalized hosts.
- **Omission of Unstructured Data:** A significant portion of a listing's appeal is driven by visual aesthetics (photos) and guest sentiment (review text). By relying solely on tabular metadata, our model treats a "renovated designer studio" and a "dated studio" identically if their location and amenities match, leaving roughly 47% of price variance unexplained.

## 7.7 Ethical Considerations

Data collection was conducted in strict adherence to ethical and legal guidelines:

- **Platform Compliance:** All scraping activities utilized the `PyAirbnb` library with enforced rate-limiting to avoid degrading service for real users. We respected the `robots.txt` directives where applicable and did not bypass authentication walls.
- **Data Minimization (GDPR):** No Personally Identifiable Information (PII) regarding hosts or guests was stored. Host names and profile images were excluded from the dataset; only the anonymous `host_id` and aggregated metrics (e.g., `host_years`) were retained for modeling.
- **Academic Use:** The derived dataset was constructed exclusively for educational analysis within the scope of this project and was not monetized or republished.

## 8 Conclusion and Future Work

This project demonstrates that while structural attributes are the foundation of real estate value, the "time" dimension is an essential component of modern dynamic pricing. By constructing a **Hybrid Dataset**—fusing historical depth with fresh, seasonal scrapings—we successfully built a Random Forest model that explains over 52% of price variance in the complex Paris market, significantly outperforming static baselines.

Our results confirm that seasonality is a "medium-impact" driver: it does not override location or size, but it is necessary for accurate valuation during peak periods.

**Future Work:** To bridge the remaining "unexplained variance" gap, future iterations of this work should focus on **Multi-Modal Learning**. Integrating Computer Vision to score listing photos (e.g., "Modernity Score") and Natural Language Processing (NLP) to analyze review sentiment would likely capture the intangible "luxury factor" that tabular data misses.